

UMC 110nm Logic and Mixed-Mode AI Advanced Enhancement 2.5V, Operated at 3.3V, NCAP SPICE Model Document

**Version 0.2 Phase 1
2012/04/25**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2012/04/25	Min Feng Jian	Origin release

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI
Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence SPECTRE version 10.1.1.111.isr8 32bit
Mentor Graphics ELDO version v6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TT: Typical NCAP case
FF: Maximum NCAP case
SS: Minimum NCAP case

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib ". /ModelFileName.lib" CaseName
, where CaseName could be TT, FF, or SS.
- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName LF=2E-6 WF=5E-6 NF=2

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation

directory.

- Add the following statement in the input file to the simulator.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be TT, FF, or SS.

- Define device condition by 'Xxx' statement.

Example:

X1 (VGG GND) DeviceName LF=2E-6 WF=5E-6 NF=2

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

- Add the following statement in the input file to the simulator.

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be TT, FF, or SS.

- Define device condition by 'Xxx' statement.

Example:

X1 VGG GND DeviceName LF=2E-6 WF=5E-6 NF=2

ModelFileName: l110ae_ncap25od33_v021

DeviceName: ncap_25od33

Geometry description (Before shrink):

$$0.25\mu\text{m} \leq \text{LF} \leq 2\mu\text{m}$$
$$2.5\mu\text{m} \leq \text{WF} \leq 10\mu\text{m}$$
$$2 \leq \text{NF} \leq 8$$

Voltage Range:

-3.3V (*1.1) <= VGS <= 3.3V (*1.1)

Temperature range:

-50°C ~ +125°C

Layout example:

The following figure is the layout example of the NCAP. LF and WF are the single poly finger length and width, respectively, and Nf is the total finger number.

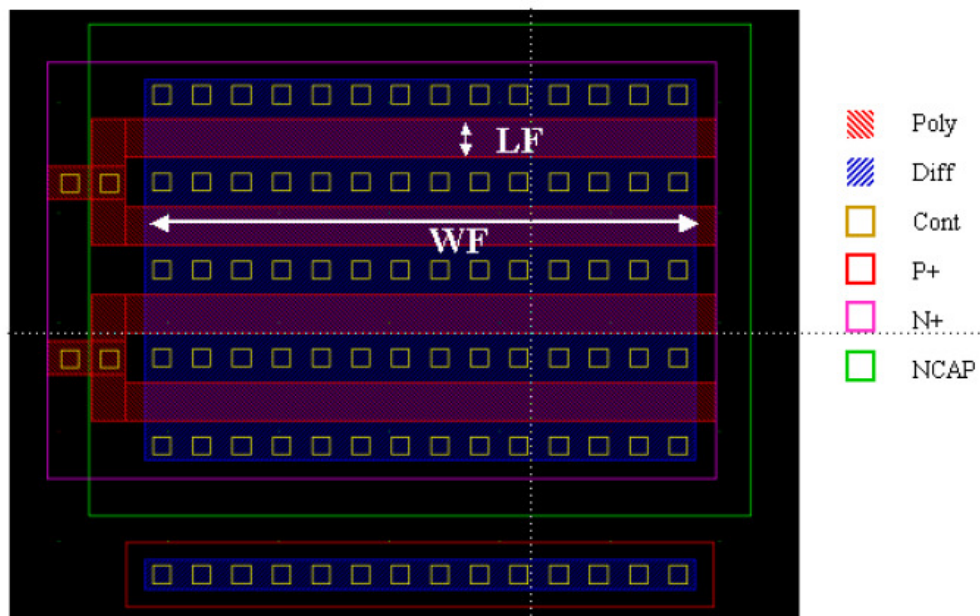


Figure 1-1 NCAP layout

2 Capacitance Model

2.1 Silicon Wafer for Modeling

The information of the hardware used for characterization and model parameter extraction is summarized in the table below.

Table 2-1 Test wafer information for NCAP

	Wafer for NCAPs measurements data
Mask ID	A11007
Lot ID	D5FHR
Wafer Number	#11

2.1.1 Test devices

The following geometries were available for parameters extraction.

Table 2-2 Geometry of NCAP devices measured for parameters extraction.

W\Lf	2	1	0.5	0.35	0.25
10			X		
5	X	X	X	X	X
2.5			X		

2.1.2 Measurement conditions

The capacitance was measured under the conditions summarized in the table below.

Table 2-3 Test conditions for Cg curves of NCAP

$V_{DD} = 3.3V$	C_G vs. V_G
V_G	From -3.63 to 3.63 V in 0.066 V step
V_B	0V
Temp [°C]	-50,25,125

2.2 Electrical Parameters

The following table shows the measured and simulated Cmax (fF/um²), where Cmax is the capacitance at VG=3.3V, VB=0 V, T=25°C.

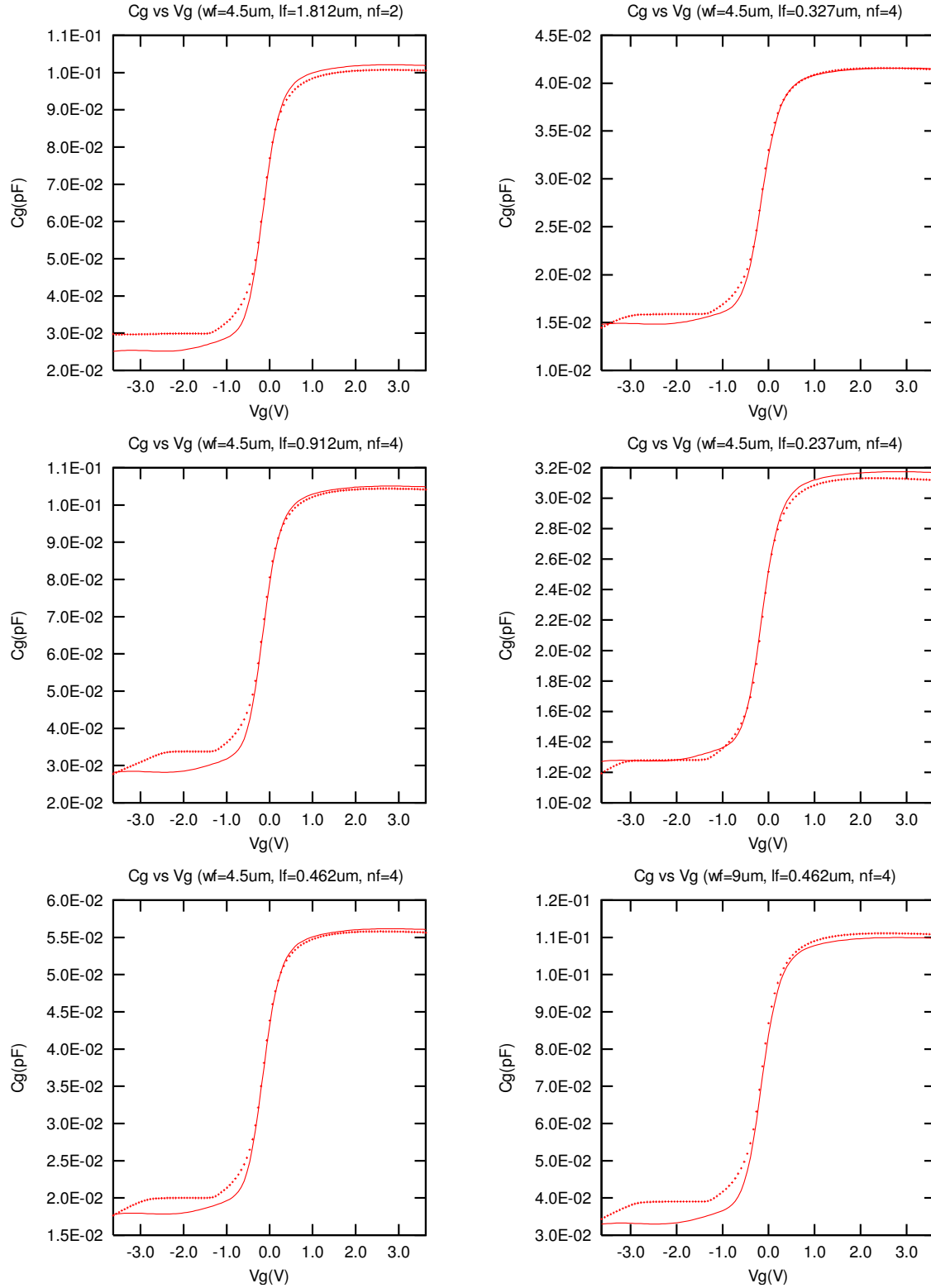
Table 2-4 Electrical Parameters

Wf*0.9 (um)	Lf*0.9+0.012 (um)	nf	Target (as- centerized) (fF/um ²)	Measured Cmax* (fF/um ²)	Model Cmax* (as-extracted) (fF/um ²)	Model Cmax* (as-centerized) (fF/um ²)
4.5	1.812	2	6.396	6.171	6.255	6.484
4.5	0.912	4		6.352	6.396	6.630
4.5	0.462	4		6.701	6.749	6.996
4.5	0.327	4		7.049	7.059	7.318
4.5	0.237	4		7.322	7.434	7.707
9	0.462	4		6.670	6.607	6.849
4.5	0.462	8		6.733	6.748	6.996
2.25	0.462	4		6.861	6.990	7.247
4.5	0.462	2		6.758	6.749	6.996

*Cmax is normalized by the area (Wf*0.9 and Lf*0.9+0.012) after shrinkage.

2.2.1 Cg vs. Vg plots

The following figures show the measured and simulated Cg (pF) vs. Vg (V) after shrinkage. The measurement data is dots and simulation is line, respectively.



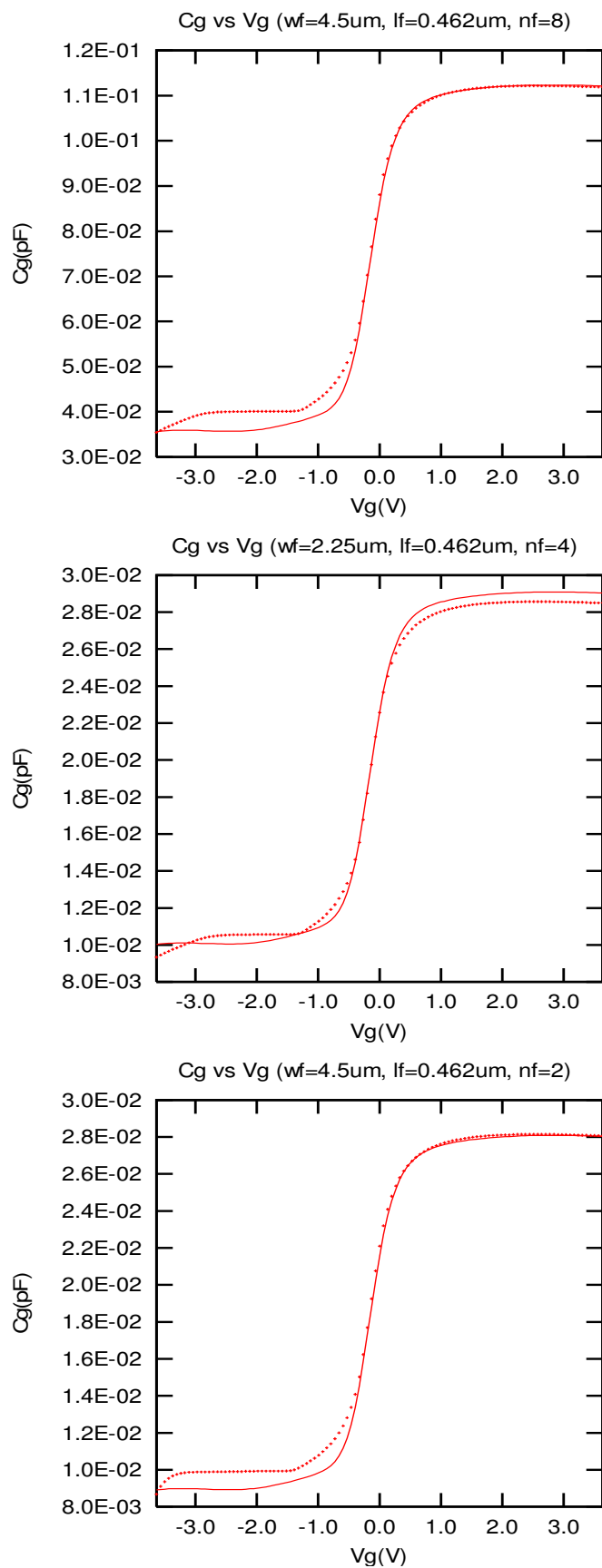
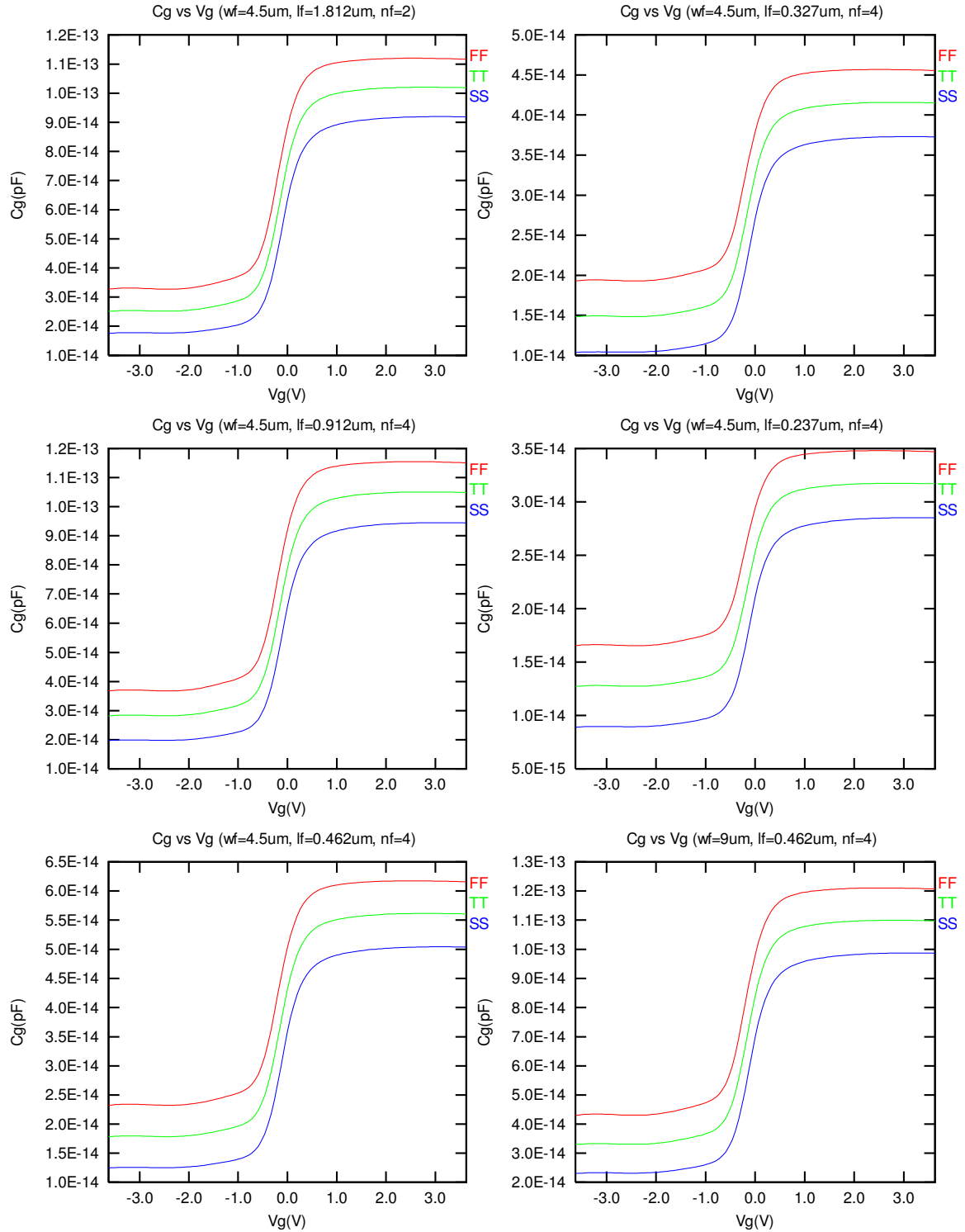


Figure 2-1 C_g vs. V_g at 25 °C

2.2.2 Corner Cg vs. Vg plots

The following figures show the simulated corner Cg (pF) vs. Vg (V) plots.



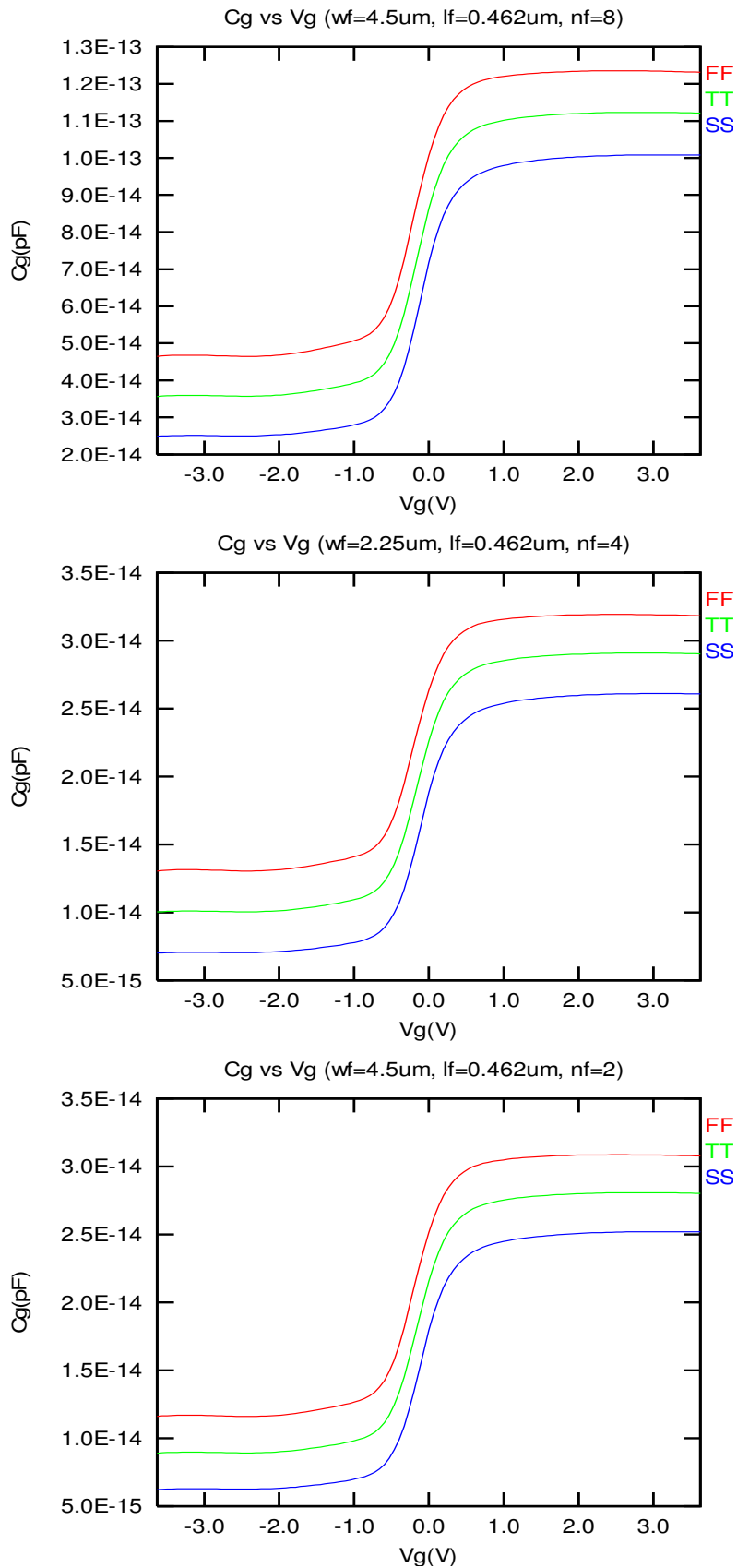


Figure 2-2 Corner Cg vs. Vg

2.3 Equation to Estimate Maximum Capacitance

The following equation can be used to estimate the capacitance value at ($V_{gb}=3.3V$) at low frequency (100kHz) for given dimension (LF, WF, NF)

$$C = 1.0367 \times 6.0 \times 10^{-15} \times (LF \times 10^6 + 0.054) \times (WF \times 10^6 + 0.036) \times NF$$

The unit for capacitance is Farad, for LF & WF is meter.

For example, for a NCAP with LF=1.812um, WF=4.5um, NF=2(after shrinkage), the capacitance value at $V_{gb}=3.3V$ is about 1.053E-13 (F).

3 Appendix

3.1 HSPICE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

3.2 SPECTRE Warning Message

Item	Warning Message	Explanation
1.	None.	None.

3.3 ELDO Warning Message

Item	Warning Message	Explanation
1.	None.	None.